



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Quiz 1 SOLUTION	Marks 30 Weight 4%	Feb 09, 2026
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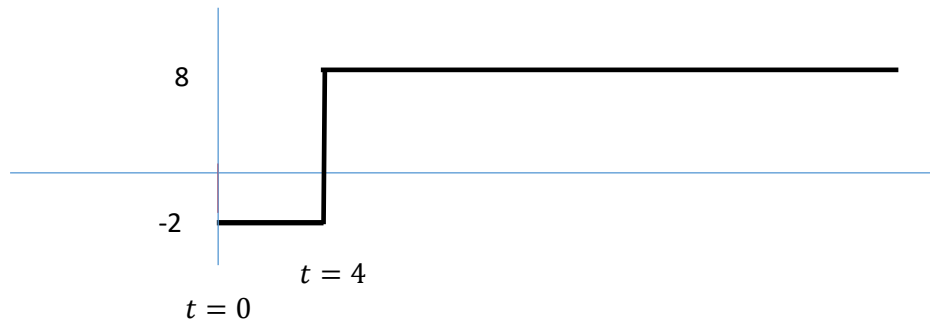
Important Notes

Clearly write your name and student ID on top.

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
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Problem 1 [10 pts]

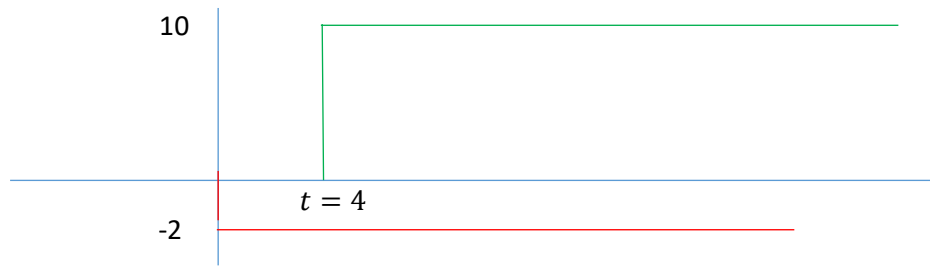
(a) Write the given signal in terms of two Unit Step Functions.



(b) Consider the complex exponential e^{st} where $s = \sigma + j\omega$ and σ, ω are real and non-zero. At what time t is $|e^{st}| = \frac{1}{e}$? (Justify your answer mathematically)

Solution:

(a) We note that graphically the required signal can be formed by two step functions as follows



Writing these out mathematically, gives

$$x(t) = 10u(t - 4) - 2u(t)$$

(b) We first note that $|e^{st}| = e^{\sigma t} |e^{j\omega t}|$, this since $e^{\sigma t}$ is a positive real number. Further, since $|e^{j\omega t}| = 1$, we have $|e^{st}| = e^{\sigma t}$. As required, we want $|e^{st}| = e^{\sigma t} = \frac{1}{e}$, which should occur at $t = -\frac{1}{\sigma}$.

Problem 2 [10 pts]

(a) Given even function $x(t)$ and odd function $y(t)$, determine whether

$$g(t) = \frac{x^3(t) + y(t)}{3}$$

Is even, odd, or neither.

(b) Is the following signal periodic? (Justify your answer)

$$x(t) = \begin{cases} \cos(2t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Solution:

(a) The signal fails both $x(t) = x(-t)$ and $x(t) = -x(-t)$ tests, and is therefore, neither even, nor odd.

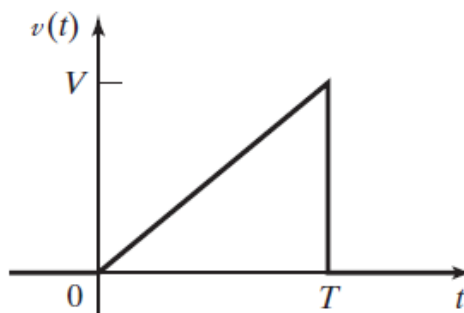
(b) The given signal is not periodic since there is no pattern that repeats itself all the time (e.g., the left half of the function is zero while right half is cosine).

Problem 3 [10 pts]

Consider the signal $v(t)$ shown below, with $V = 10$ and $T = 10$. Evaluate the following

(a) $\int_{-\infty}^{\infty} v(t)\delta(t - 15)dt$

(b) $\int_{-\infty}^{\infty} v(t)u(t)dt$

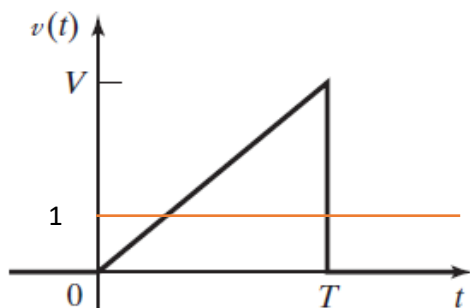


Solution:

(a) By sampling property of impulse, we have

$$\int_{-\infty}^{\infty} v(t)\delta(t - 15)dt = v(15) = 0$$

(b) The required integral is the area under the product of unit step and $v(t)$. Let's draw the two function to see their product (unit step in red, $v(t)$ in black)



Clearly the product of the two functions is again $v(t)$. So we only need to find the area under $v(t)$. Which is easily seen to be $\frac{VT}{2} = \frac{10 \times 10}{2} = 50$.